SOFTWARE TESTING ASSIGNMENT

E-Commerce Website Test Plan & RTM

Demo application: SauceDemo

Website: https://www.saucedemo.com/

Document purpose
This document defines the testing scope, objectives, approach, entry and exit criteria, resources, schedule, risks, traceability, and deliverables for a demo e-commerce application.

Prepared for: Software Testing Assignment

Document version: 1.0 | Date: 29 October 2026

1. Overview

This test plan covers the core shopping journey on SauceDemo, a demo web application commonly used to practise e-commerce testing. The plan is intended for educational testing only. Any test data, credentials, or behavior should be verified against the current demo environment before execution.

1.1 Application under test

Application: SauceDemo demo e-commerce website
URL: https://www.saucedemo.com/
Primary user journey: Login → Browse products → Add products to cart → Review cart → Enter checkout information → Complete checkout → View confirmation.

1.2 Test plan objectives

• Verify that critical shopping features work according to expected behavior.
• Identify functional defects before the application is considered ready for demonstration.
• Check validation, navigation, and consistency across the shopping flow.
• Map requirements to test cases and execution evidence using an RTM.
• Record risks, assumptions, and completion criteria so testing can be tracked consistently.

2. Scope

2.1 In-scope features

ID	Feature	Coverage
IS-01	Login / logout	Valid and invalid login, required fields, logout behavior.
IS-02	Product catalogue	Product list display, product names/prices, product details, sorting if available.
IS-03	Shopping cart	Add/remove items, cart badge/count, cart contents and navigation.
IS-04	Checkout information	Required-field validation and handling of customer information.
IS-05	Order overview	Items, quantities, subtotal, tax, and total display where supported.
IS-06	Order completion	Complete checkout and verify confirmation message/page.
IS-07	Basic usability	Navigation, visible labels, buttons, and obvious layout issues on desktop browser.

2.2 Out-of-scope features

• Real payment gateway processing, refunds, and financial transactions.
• Real shipping, inventory, order fulfilment, and email/SMS delivery.
• Security penetration testing, vulnerability scanning, and load/stress testing.
• Backend database verification or source-code testing where access is not provided.
• Production data, real customer accounts, and third-party integrations.
• Exhaustive testing on every browser, device, language, or accessibility standard.

3. Test approach and methodology

Testing will primarily be manual and requirement-based. A tester will prepare test cases, execute them in a supported desktop browser, record actual results, and raise defects for unexpected behavior. Testing should proceed from high-priority critical paths to lower-priority checks.

3.1 Test levels and types

• Smoke testing: confirm the site loads and the main shopping journey is testable.
• Functional testing: validate login, product selection, cart, checkout, and confirmation.
• Negative testing: enter invalid credentials or leave required checkout fields blank.
• Integration-flow testing: verify data is carried correctly from product selection through checkout.
• Regression testing: rerun key existing flows after a fix or change.
• Usability checks: confirm labels, controls, and navigation are understandable.

3.2 Test design and execution

1. Review the requirements and identify test conditions.
2. Prepare test cases with preconditions, steps, expected results, and priority.
3. Confirm the test environment and test account/data.
4. Execute smoke tests, then functional and negative tests.
5. Record Pass, Fail, Blocked, or Not Run for each test case.
6. Log defects with steps to reproduce, expected/actual result, severity, and evidence.
7. Retest fixes and run relevant regression tests.
8. Prepare a test summary and document known limitations.

3.3 Test environment and data

• Environment: current demo site at https://www.saucedemo.com/.
• Browser: latest stable Google Chrome on a desktop/laptop; note the exact version at execution time.
• Network: stable internet connection.
• Data: demo-provided test credentials and non-sensitive sample checkout information, as supported by the site. Do not enter real payment or personal data.
• Evidence: screenshots or screen captures for failed tests and important checkpoints.

4. Entry and Exit Criteria

4.1 Entry criteria

• Test scope and core requirements have been reviewed.
• Demo website is accessible and the test environment is available.
• Test credentials and permitted sample data are known.
• Test cases and expected results are prepared and reviewed.
• No known environment issue prevents execution of the planned critical path.

4.2 Exit criteria

• All planned high-priority test cases have been executed, or any blocked/not-run cases are documented with reasons.
• All critical and high-severity defects are fixed and retested, or explicitly documented as accepted limitations.
• Relevant regression tests have passed after fixes.
• RTM is updated with test case IDs and execution status.
• A test summary report lists coverage, pass/fail/block counts, open defects, and known risks.

5. Resources and Responsibilities

Resource / Role	Responsibility
Test lead / coordinator	Confirm scope, priorities, schedule, defect triage, and final summary.
QA tester	Design and execute test cases, capture evidence, log defects, update RTM.
Developer / support contact	Investigate defects and provide fixes or environment guidance when available.
Desktop/laptop and browser	Access the demo website and run tests.
Test management record	Spreadsheet or document to track test cases, results, defects, and traceability.

6. Proposed Test Schedule

The schedule below is a sample plan for a small educational assignment. Dates can be adjusted to match the actual submission timeline.

Day	Activity	Output
Day 1	Review scope, requirements, and test environment.	Requirement list and test plan.
Day 2	Create test cases and map them to requirements.	Test cases and initial RTM.
Day 3	Execute smoke, login, catalogue, and cart tests.	Execution results and defect log.
Day 4	Execute checkout, negative, and usability tests.	Updated results and evidence.
Day 5	Retest fixes if available, run regression checks, and close documentation.	Final RTM and test summary.

7. Risks and Mitigation

Risk	Impact	Mitigation
Demo website is unavailable or slow.	Testing may be delayed or blocked.	Check access before execution; record downtime and retry later.
Demo behavior or credentials change.	Test steps may no longer work as written.	Verify current instructions and update test data before execution.
Requirements are incomplete or ambiguous.	Expected results may be interpreted inconsistently.	Document assumptions and confirm expected behavior where possible.
Limited time for full coverage.	Lower-priority cases may remain untested.	Prioritize login, cart, checkout, and order confirmation.
No access to backend or source code.	Backend and security behavior cannot be verified directly.	Limit conclusions to observable UI behavior and list this as a scope constraint.
A defect is found late in testing.	Retesting and regression may not fit the schedule.	Reserve the final day for critical retesting and document unresolved defects.

8. Requirement Traceability Matrix (RTM)

The RTM links each requirement to one or more test cases. Test case IDs below are proposed identifiers; execution status should be filled in after running the tests.

Req. ID	Requirement	Test Case ID(s)	Expected result / coverage	Priority
REQ-01	User can log in with valid demo credentials.	TC-001	User reaches the product catalogue after successful login.	High
REQ-02	Invalid login details are handled correctly.	TC-002, TC-003	Login is rejected and a clear error is shown; user remains unauthenticated.	High
REQ-03	User can view the product catalogue and product information.	TC-004, TC-005	Products, names, prices, and available details display correctly.	High
REQ-04	User can add a product to the cart and remove it.	TC-006, TC-007	Cart count and contents update correctly after add/remove actions.	High
REQ-05	User can proceed to checkout and required information is validated.	TC-008, TC-009	Checkout form accepts valid sample data and flags missing required fields.	High
REQ-06	Order overview shows correct selected items and totals.	TC-010	Selected products and displayed totals are consistent with the cart and site calculations.	High
REQ-07	User can complete checkout and see confirmation.	TC-011	Order completion page/message is displayed after valid checkout.	High
REQ-08	User can log out of the application.	TC-012	User is returned to the login page or logged-out state.	Medium

RTM status: Not Run (planned). Update each requirement with the actual result after test execution. If a requirement fails, link the defect ID in the test execution record.

9. Test Case Inventory (Summary)

Test Case ID	Scenario	Expected result
TC-001	Login using valid demo credentials.	Login succeeds and catalogue is displayed.
TC-002	Login using an invalid password.	Login is rejected with an error message.
TC-003	Submit login with a required field blank.	Validation or an error message is displayed.
TC-004	Open the product catalogue.	Product list and key product details are visible.
TC-005	Open a product detail page, if supported.	Selected product details match the selected item.
TC-006	Add a product to the cart.	Cart count and cart contents reflect the added product.
TC-007	Remove a product from the cart.	Removed product no longer appears in the cart.
TC-008	Proceed to checkout and enter valid sample information.	User can continue to the order overview.
TC-009	Leave a required checkout field blank.	Checkout is blocked and validation is displayed.
TC-010	Review order overview.	Items and displayed subtotal/tax/total are consistent.
TC-011	Complete checkout.	Order confirmation is displayed.
TC-012	Log out.	User returns to a logged-out state.

10. Test Deliverables

• Test Plan: scope, objectives, approach, criteria, resources, schedule, and risks.
• Requirement list / reviewed assumptions: requirements used to derive test coverage.
• Test cases and test data: steps, preconditions, expected results, and priorities.
• Requirement Traceability Matrix (RTM): requirement-to-test-case mapping and execution status.
• Test execution report: pass/fail/blocked/not-run status and execution notes.
• Defect log: defect ID, summary, steps to reproduce, expected/actual results, severity, status, and evidence.
• Screenshots or other evidence: supporting proof for failures and important test outcomes.
• Test summary report: overall coverage, results, open defects, limitations, and recommendation.

11. Assumptions and Limitations

• SauceDemo is used as a demonstration environment, not a production store.
• The exact credentials, product behavior, and page labels should be checked at the time of execution.
• This plan defines intended testing; it does not claim that the listed test cases have already been executed.
• No real purchases or payments should be made.
• RTM and test case statuses remain “Not Run” until actual execution evidence is recorded.

12. Conclusion

This plan provides a structured approach to checking the core e-commerce journey, from login through order confirmation. The RTM connects each requirement to test cases, while the risk register and defined entry/exit criteria help make the testing process measurable and transparent. The plan should be updated with actual results, defect IDs, and evidence after execution.